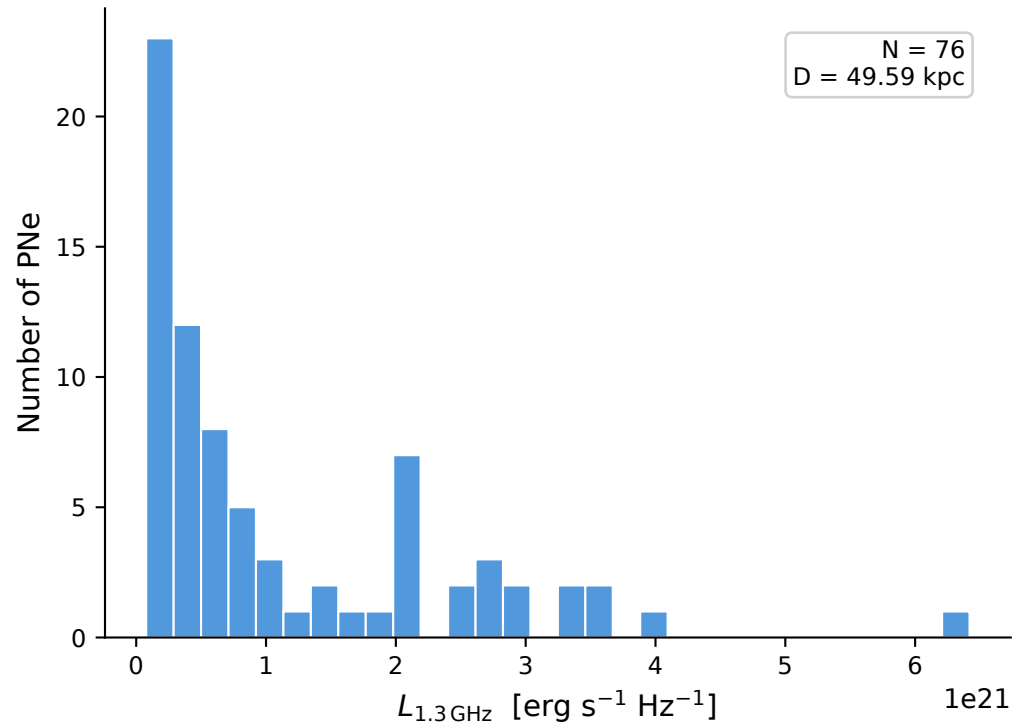
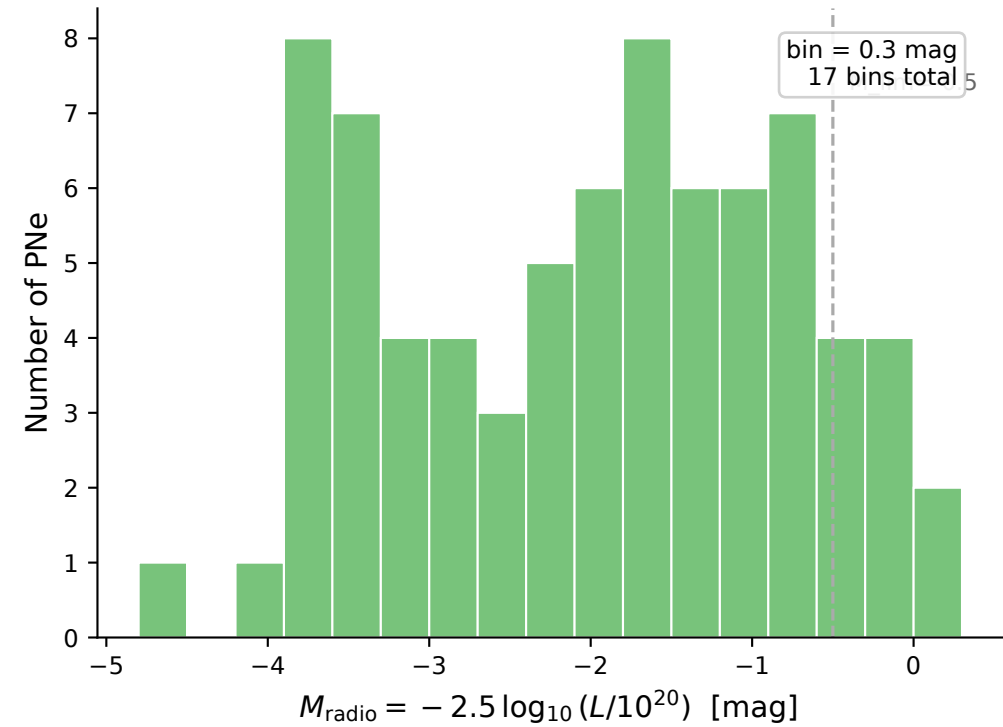


Data Preparation: From Luminosity to Fit-Ready PNLF

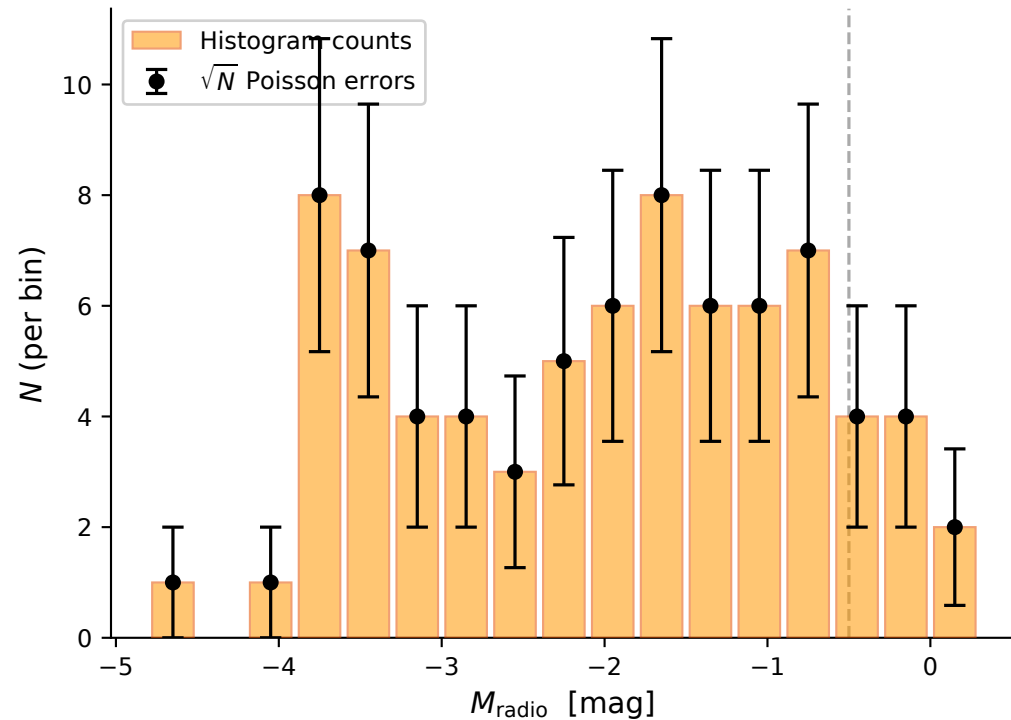
(a) Luminosity distribution



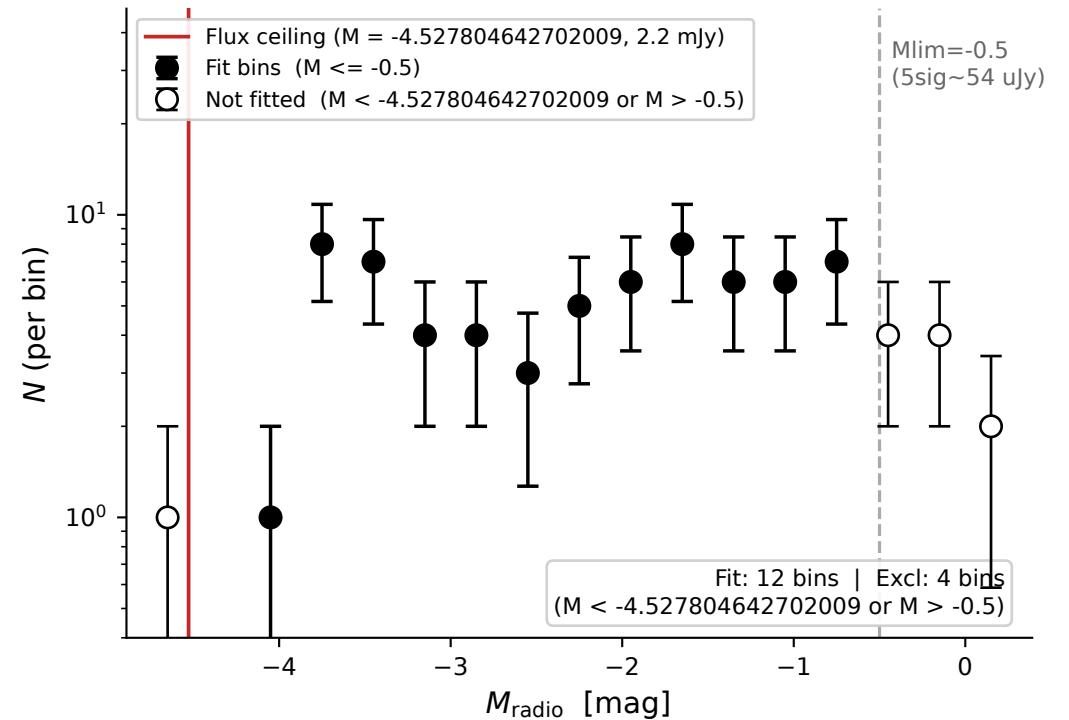
(b) Magnitude distribution

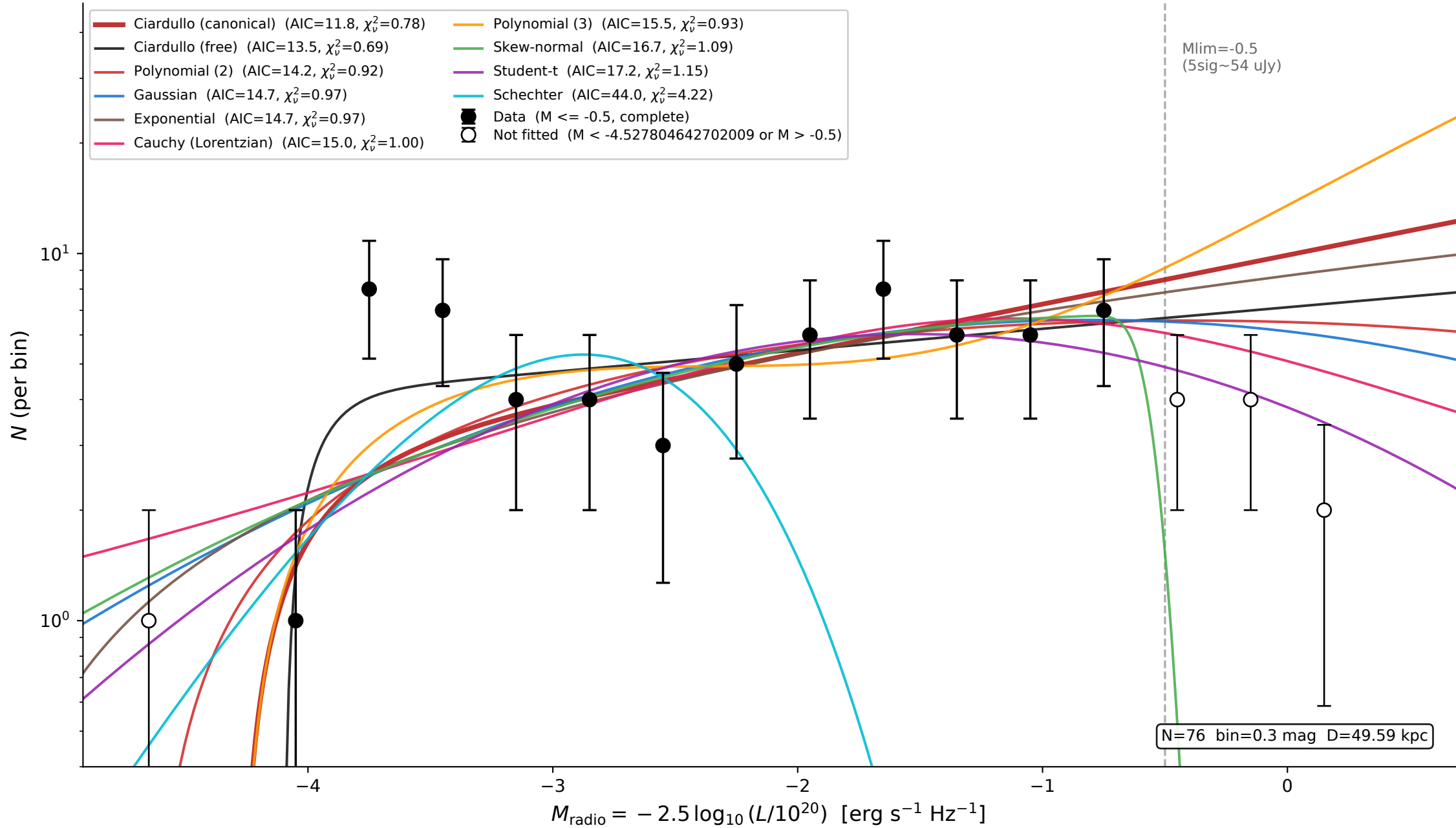


(c) Binned counts with Poisson uncertainties

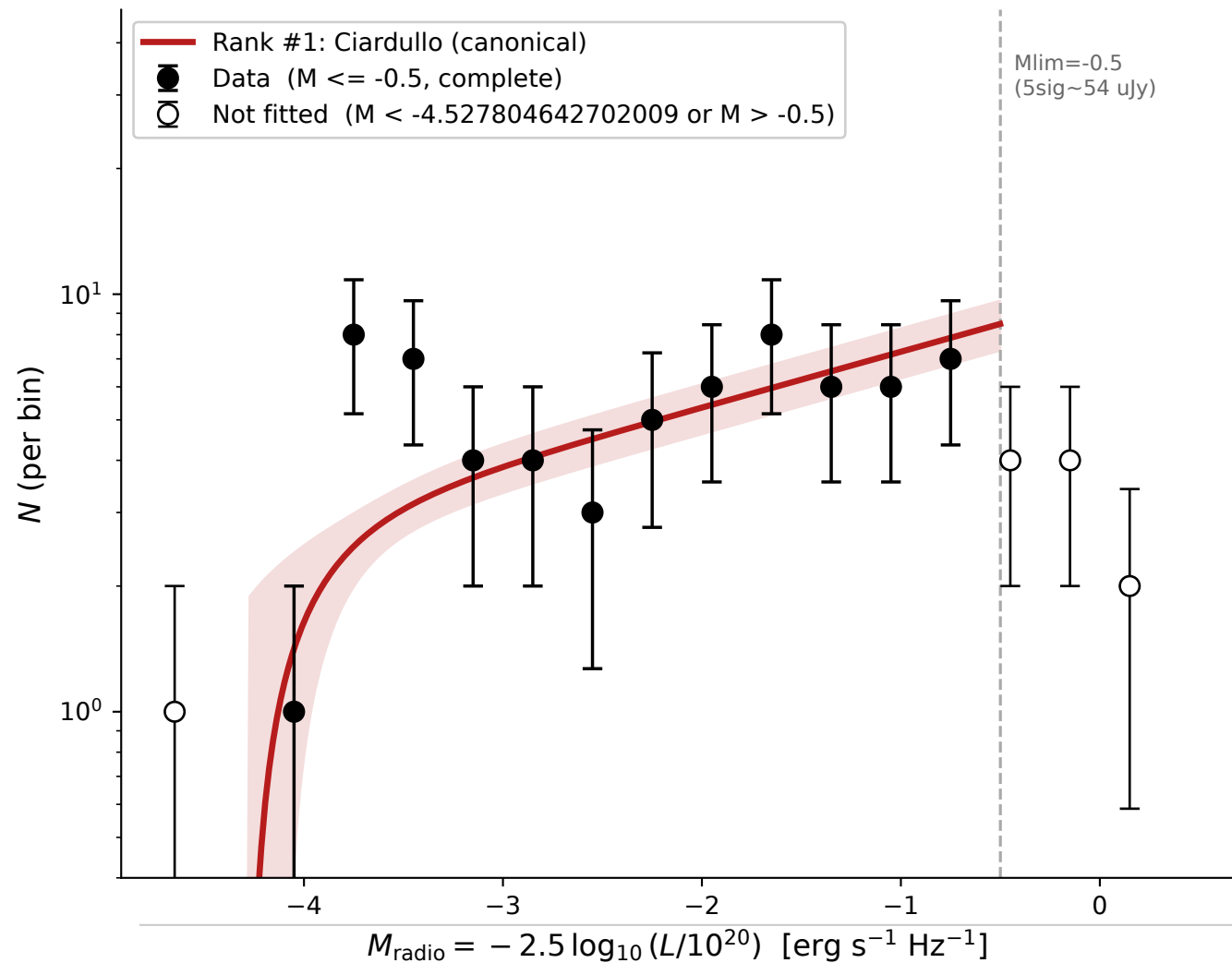


(d) Fit-ready data: complete vs excluded





Rank #1: Ciardullo (canonical)



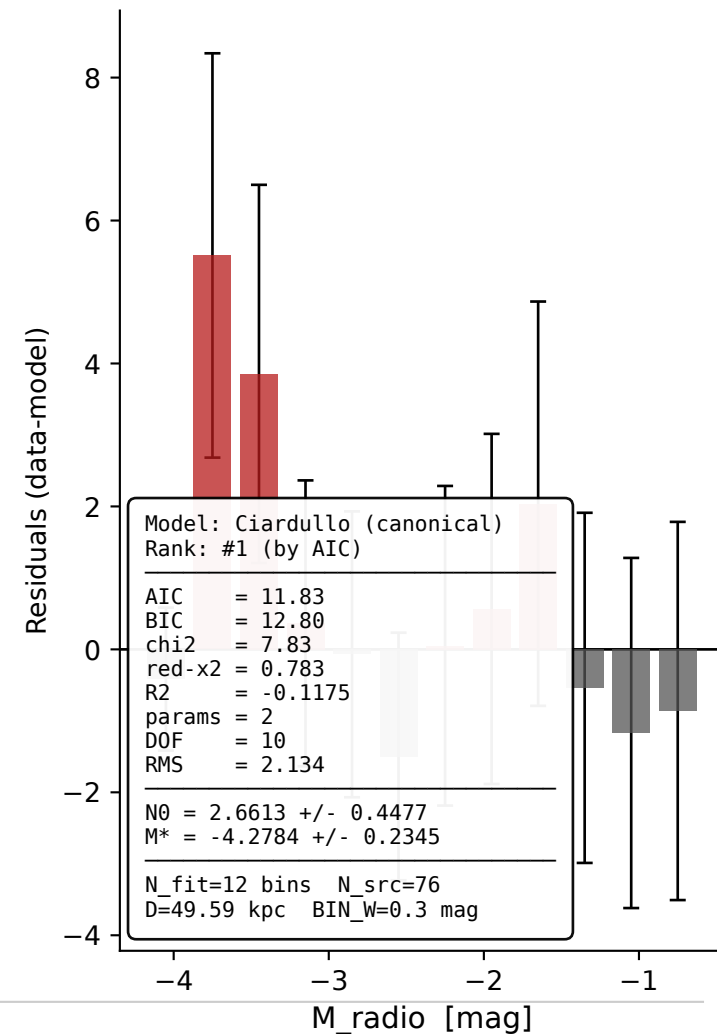
General model equation:

$N(M) = N_0 * \exp(0.307 * (M - M^*)) * (1 - \exp(3.0 * (M^* - M)))$
 $\alpha = 0.307$, $\beta = 3.0$ (FIXED – Ciardullo 1989 optical values)
 Only N_0 and M^* are free ($k=2$) – most parsimonious Ciardullo form

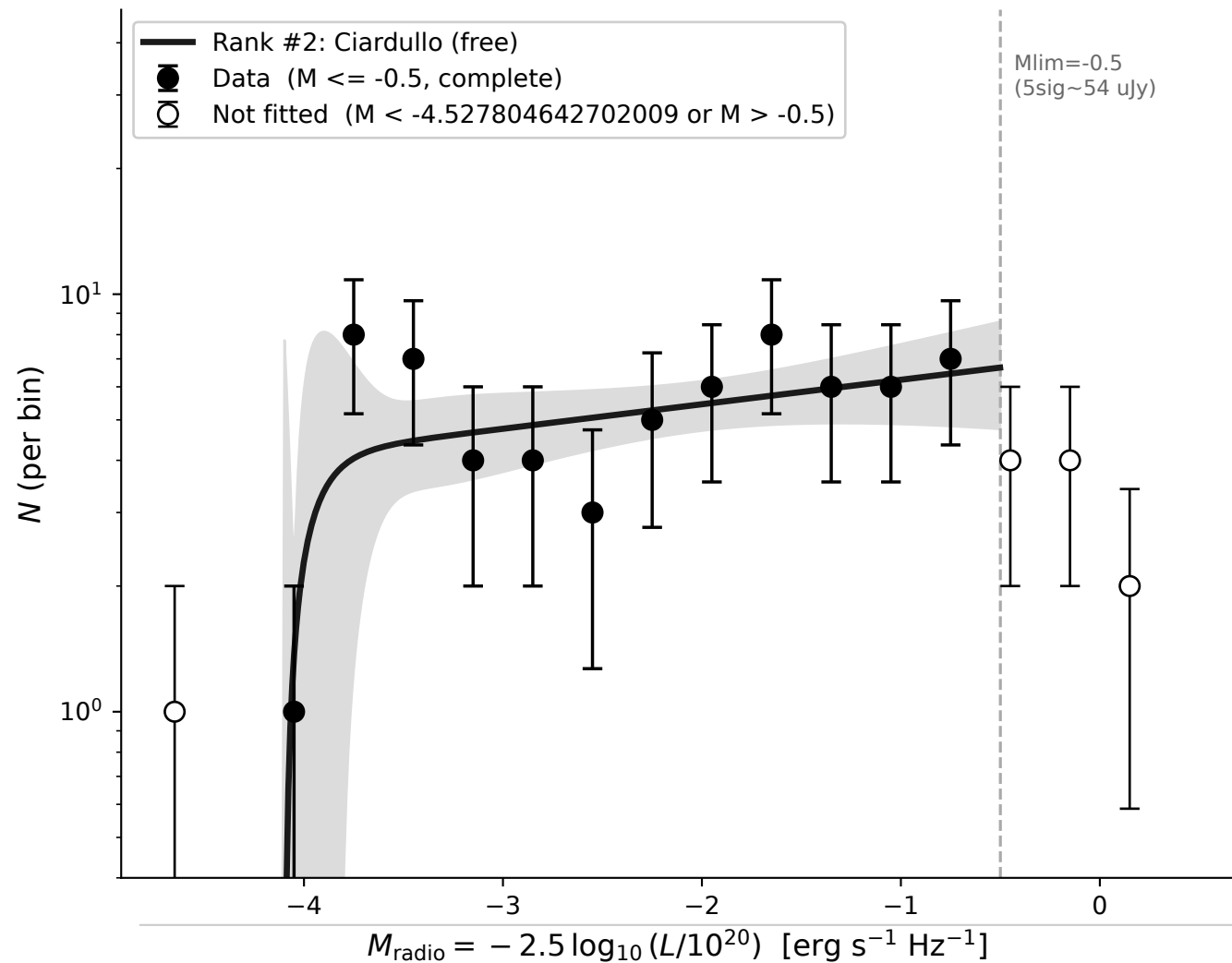
Fitted equation (parameter values substituted):

$N(M) = 2.66 * \exp(0.307 * (M - (-4.278))) * (1 - \exp(3.0 * (-4.278 - M)))$
 $N_0=2.66$ $M^*=-4.278$ $\alpha=0.307$ (fixed) $\beta=3.0$ (fixed)
 $L^* = 10^{21.711} \text{ erg/s/Hz}$

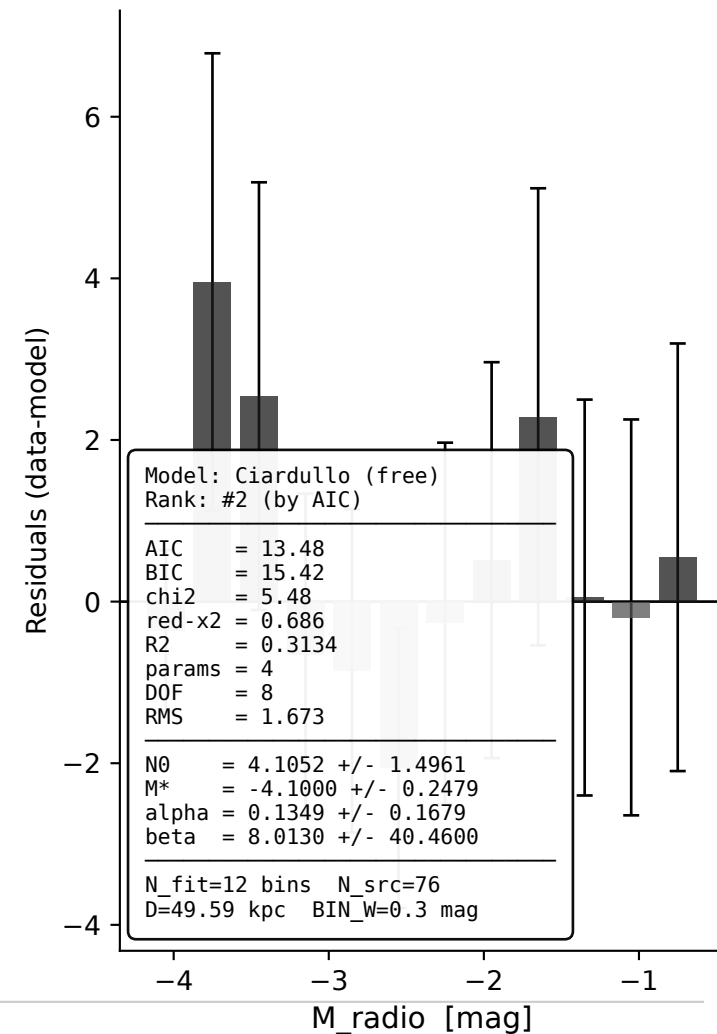
Residuals



Rank #2: Ciardullo (free)



Residuals



General model equation:

$$N(M) = N_0 * \exp(\alpha * (M - M^*)) * (1 - \exp(\beta * (M^* - M)))$$

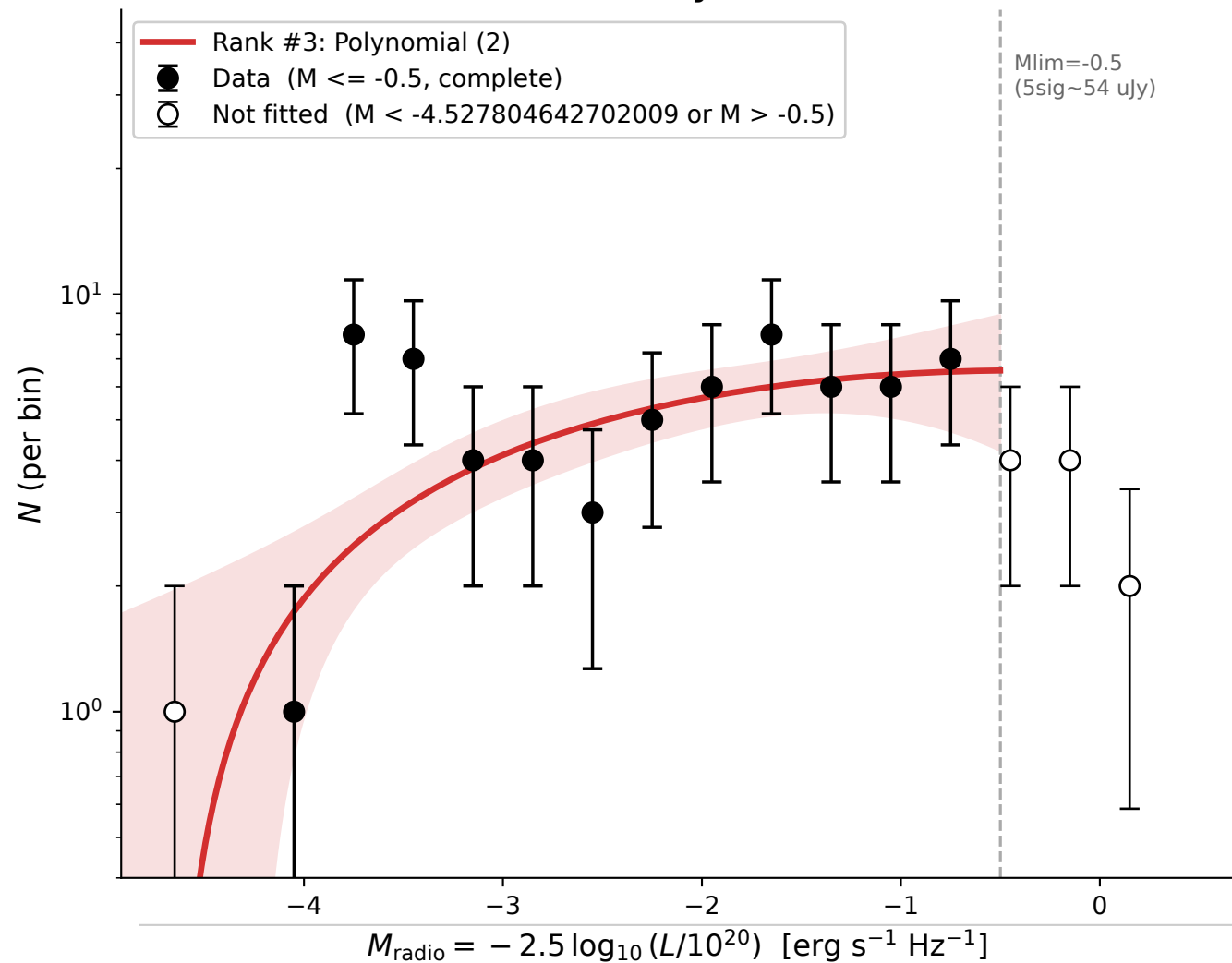
ALL parameters free: N_0 , M^* , α (faint-end slope), β (bright cutoff)
Ciardullo (1989) form – radio calibration, not fixed to optical values

Fitted equation (parameter values substituted):

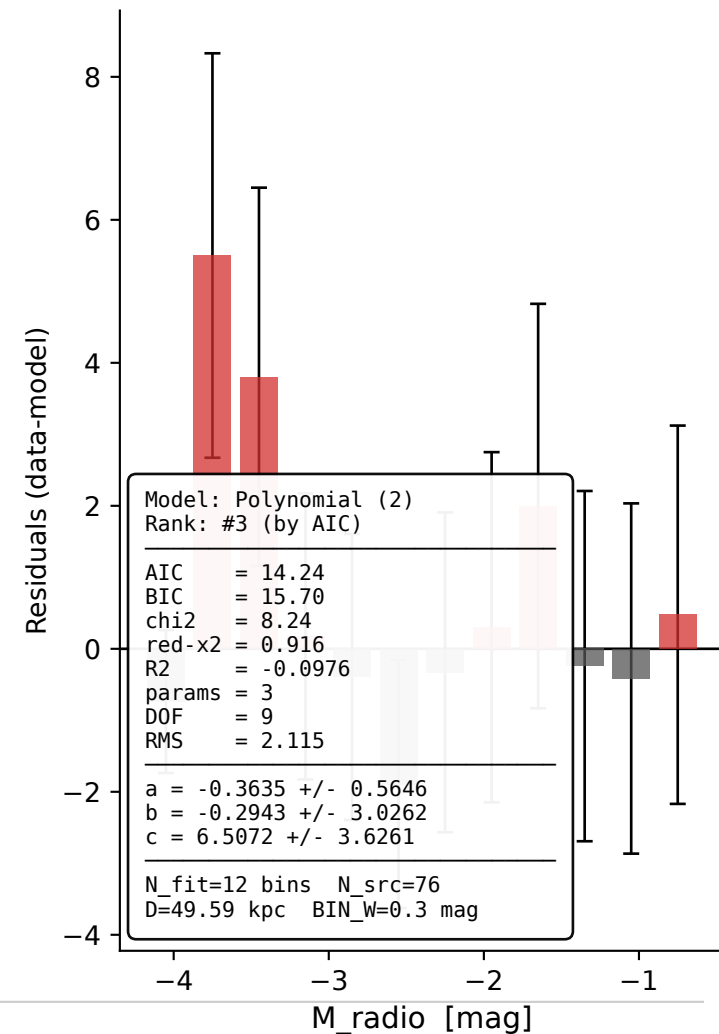
$$N(M) = 4.11 * \exp(0.135 * (M - (-4.100))) * (1 - \exp(8.013 * (-4.100 - M)))$$

 $N_0=4.11$ $M^*=-4.100$ $\alpha=0.135$ $\beta=8.013$
 $L^* = 10^{21.640} \text{ erg/s/Hz}$

Rank #3: Polynomial (2)



Residuals



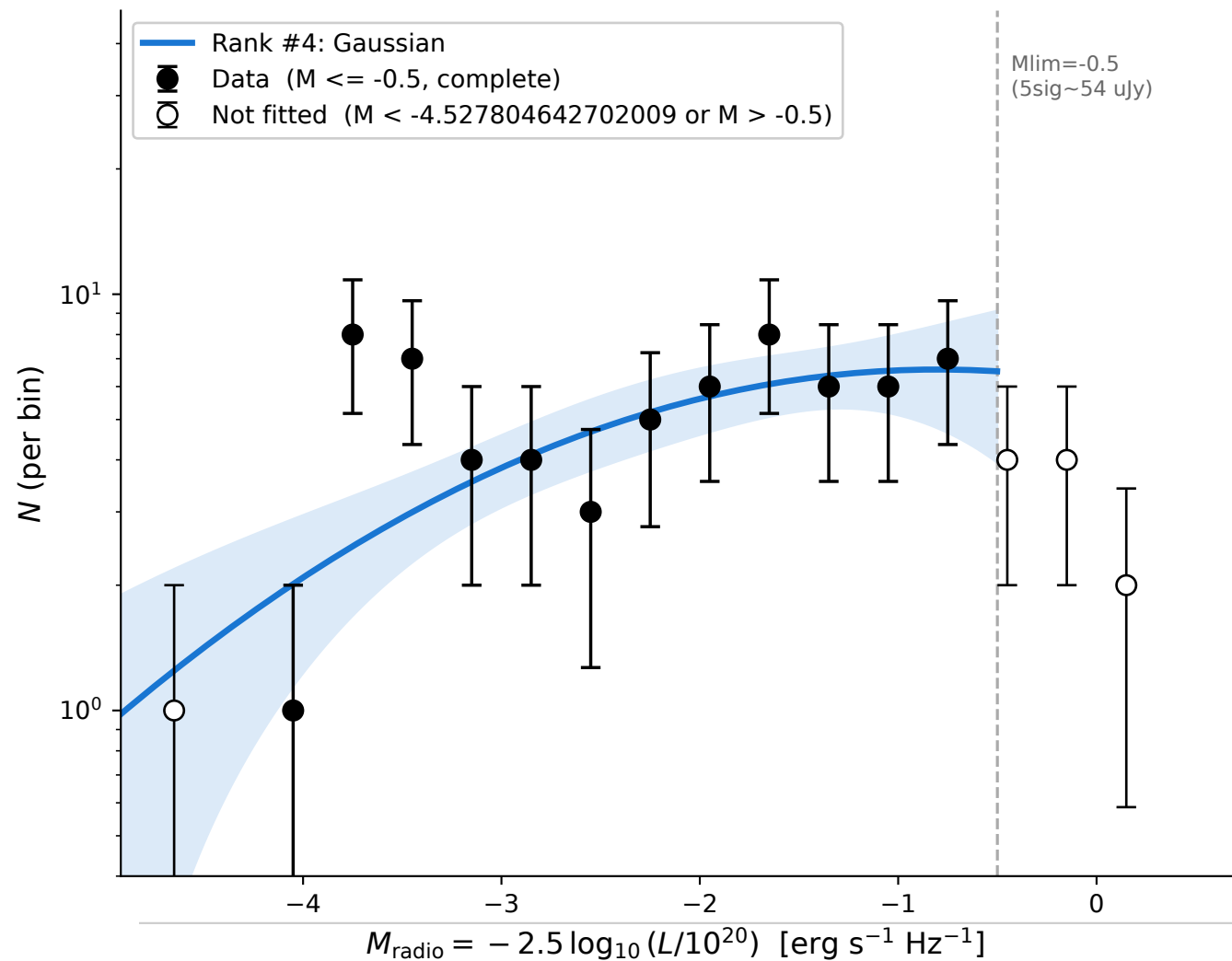
General model equation:

$$N(M) = a \cdot M^2 + b \cdot M + c$$

Fitted equation (parameter values substituted):

$$N(M) = -0.3635 \cdot M^2 - 0.2943 \cdot M + 6.5072$$

Rank #4: Gaussian



General model equation:

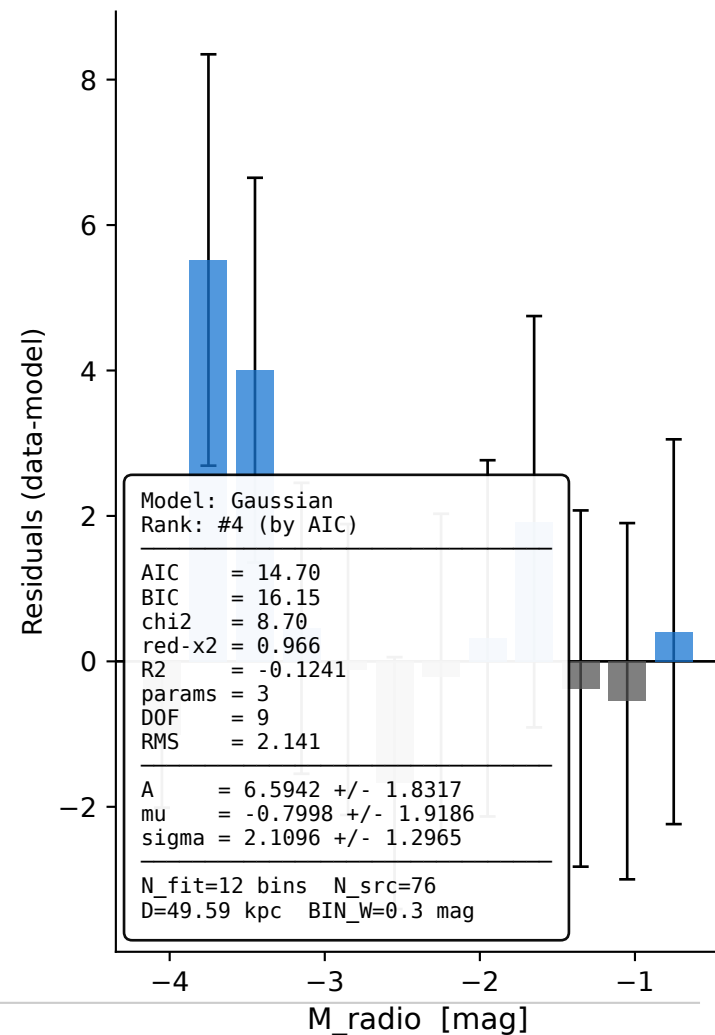
$$N(M) = A * \exp[-0.5 * ((M - \mu) / \sigma)^2]$$

A = amplitude, μ = peak magnitude, σ = width

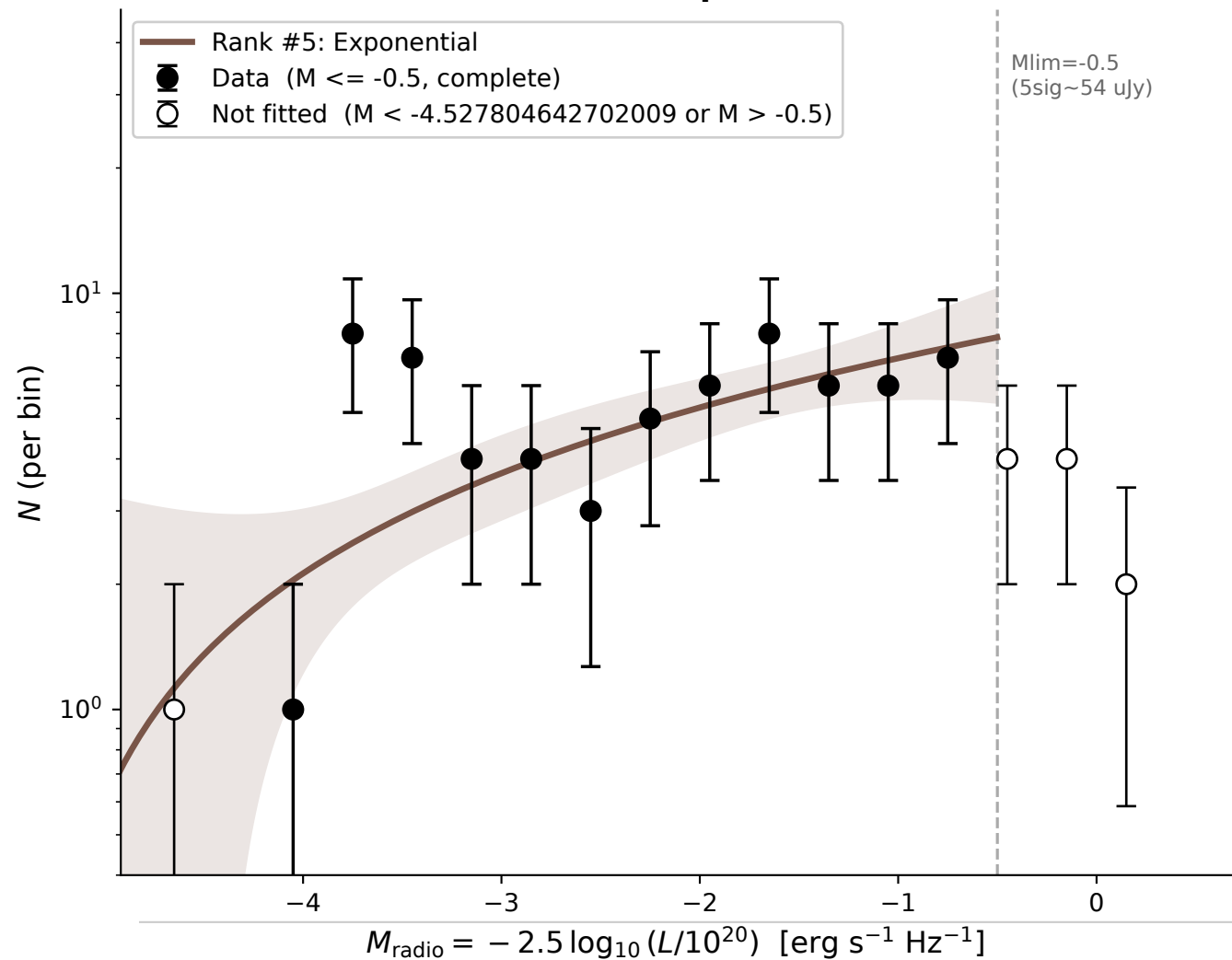
Fitted equation (parameter values substituted):

$$N(M) = 6.59 * \exp[-0.5 * ((M - (-0.800)) / 2.110)^2]$$

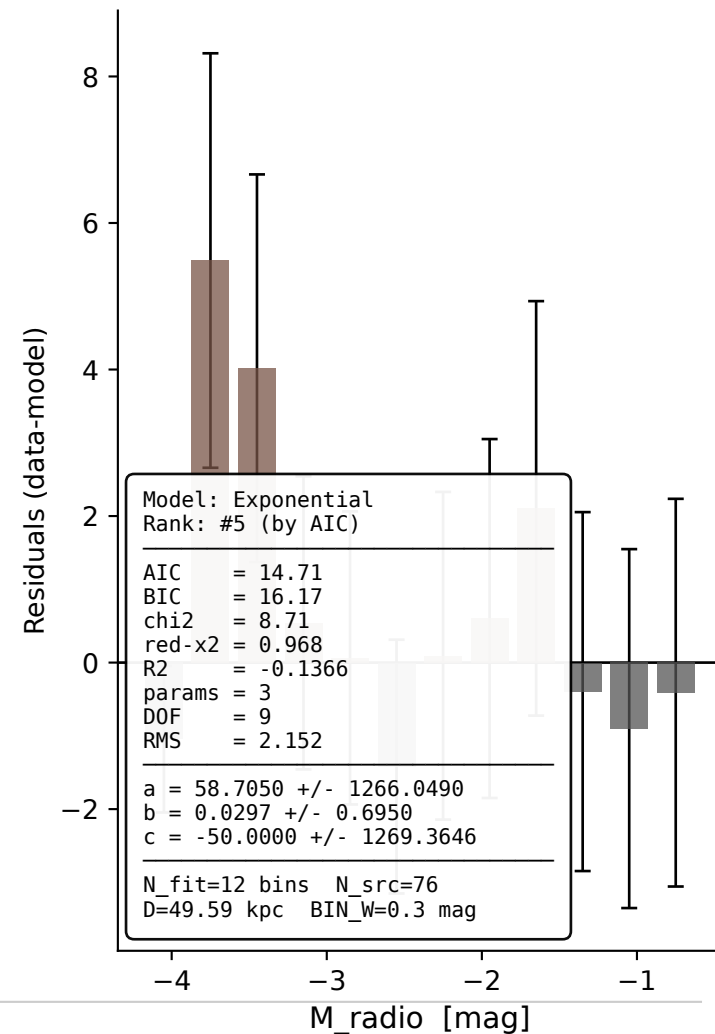
Residuals



Rank #5: Exponential



Residuals



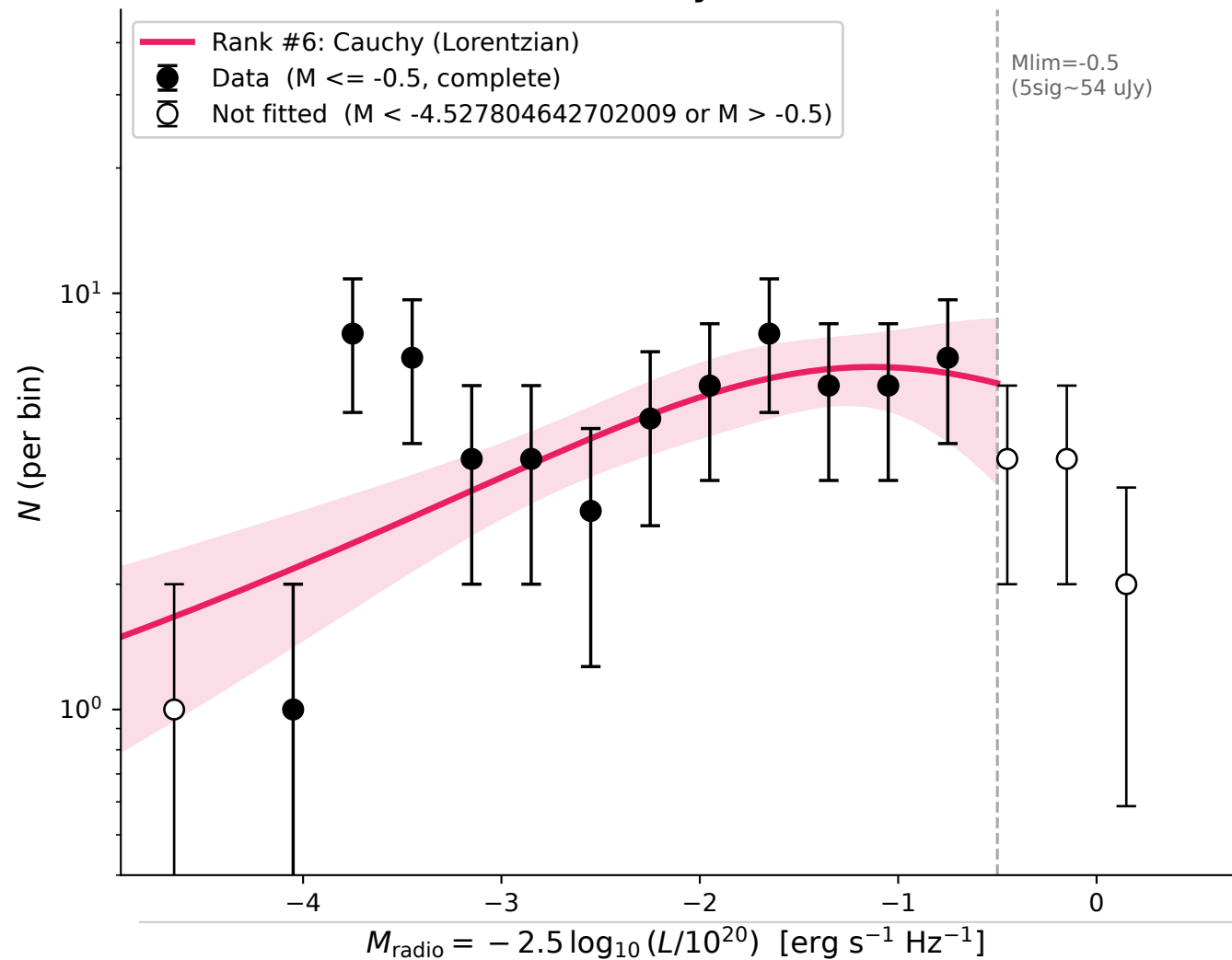
General model equation:

$N(M) = a * \exp(b * M) + c$
 Simple exponential rise toward faint magnitudes
 a = amplitude, b = rate (>0 = rising toward faint), c = offset

Fitted equation (parameter values substituted):

$N(M) = 58.7050 * \exp(0.0297 * M) - 50.0000$

Rank #6: Cauchy (Lorentzian)



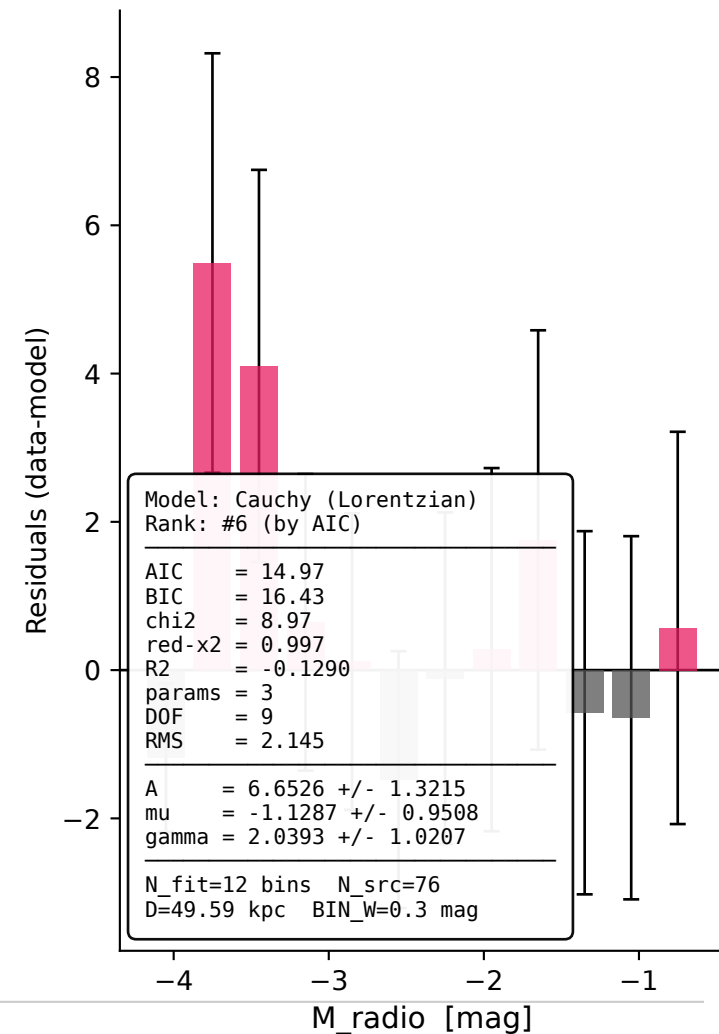
General model equation:

$$N(M) = A / [1 + ((M - \mu) / \gamma)^2]$$
 Heavy-tailed symmetric distribution (heavier tails than Gaussian)
 A = peak height, μ = centre, γ = half-width at half-maximum

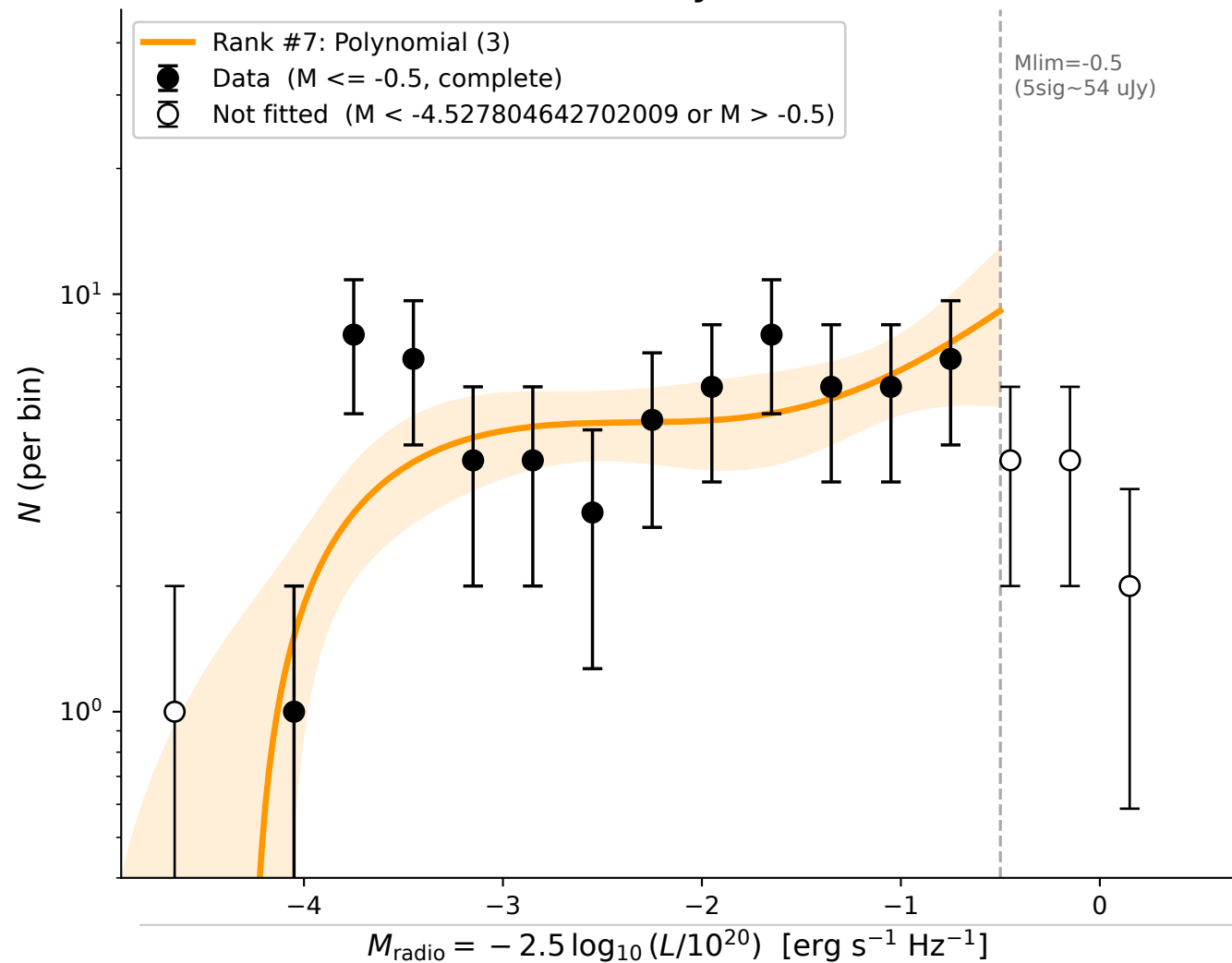
Fitted equation (parameter values substituted):

$$N(M) = 6.65 / [1 + ((M - (-1.129)) / 2.039)^2]$$

Residuals



Rank #7: Polynomial (3)



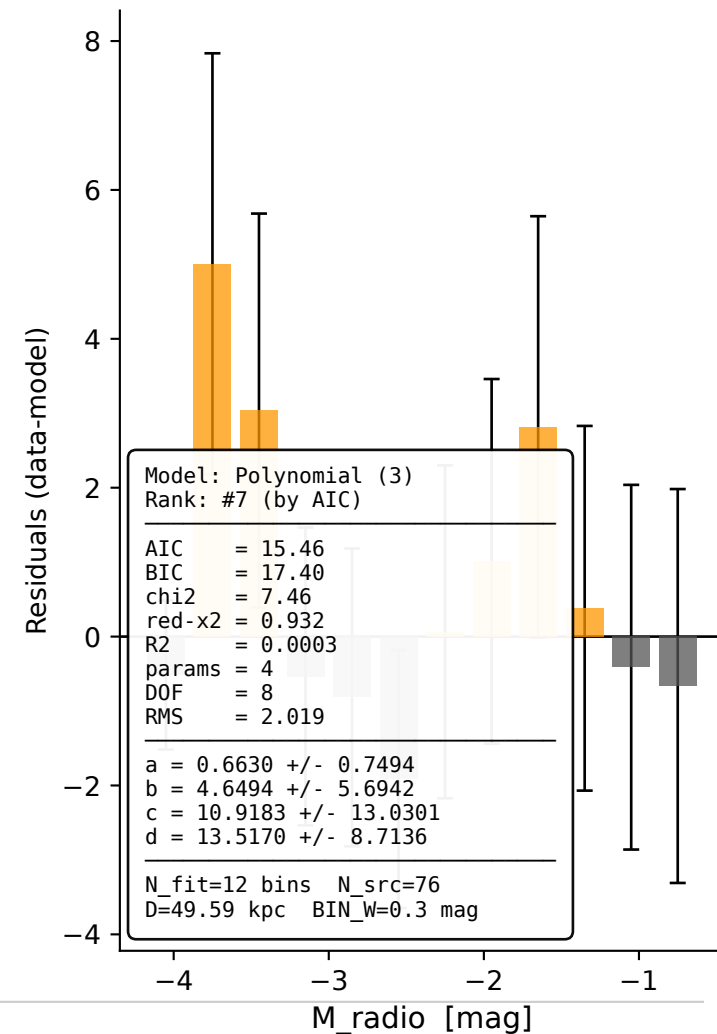
General model equation:

$$N(M) = a \cdot M^3 + b \cdot M^2 + c \cdot M + d$$

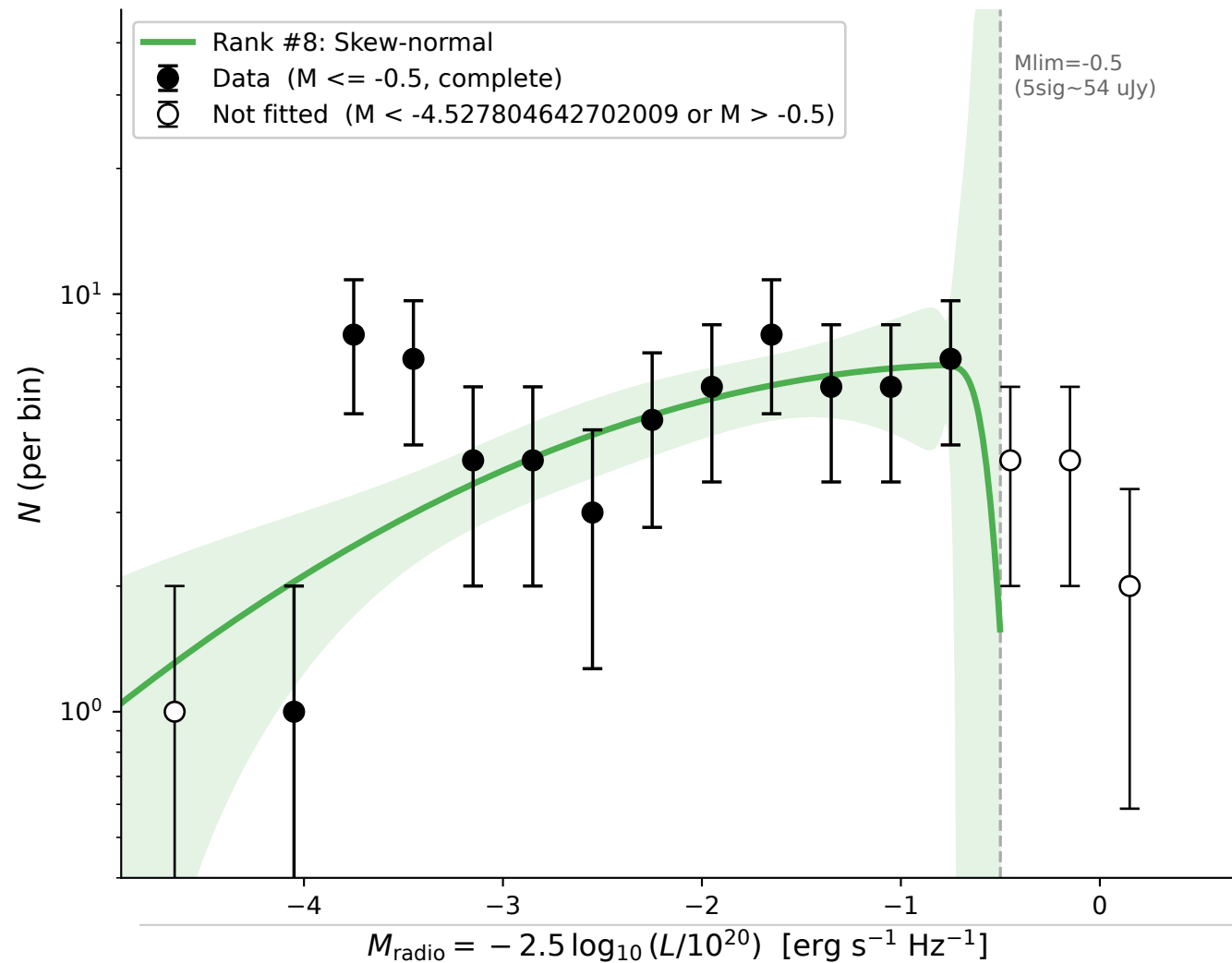
Fitted equation (parameter values substituted):

$$N(M) = 0.6630 \cdot M^3 + 4.6494 \cdot M^2 + 10.9183 \cdot M + 13.5170$$

Residuals



Rank #8: Skew-normal



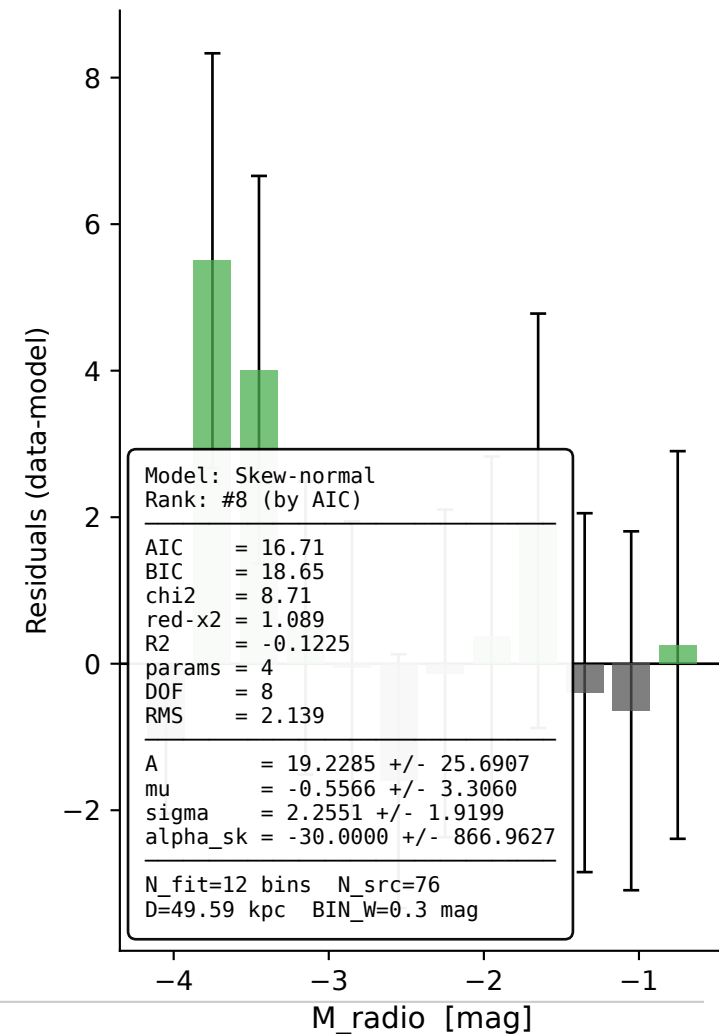
General model equation:

$N(M) = A * f_{\text{SN}}(M; \mu, \sigma, \alpha_{\text{sk}})$
 $f_{\text{SN}}(x) = (2/\sigma) * \phi((x-\mu)/\sigma) * \Phi(\alpha_{\text{sk}} * (x-\mu)/\sigma)$
 ϕ = normal PDF, Φ = normal CDF, $\alpha_{\text{sk}} < 0$ = left-skewed

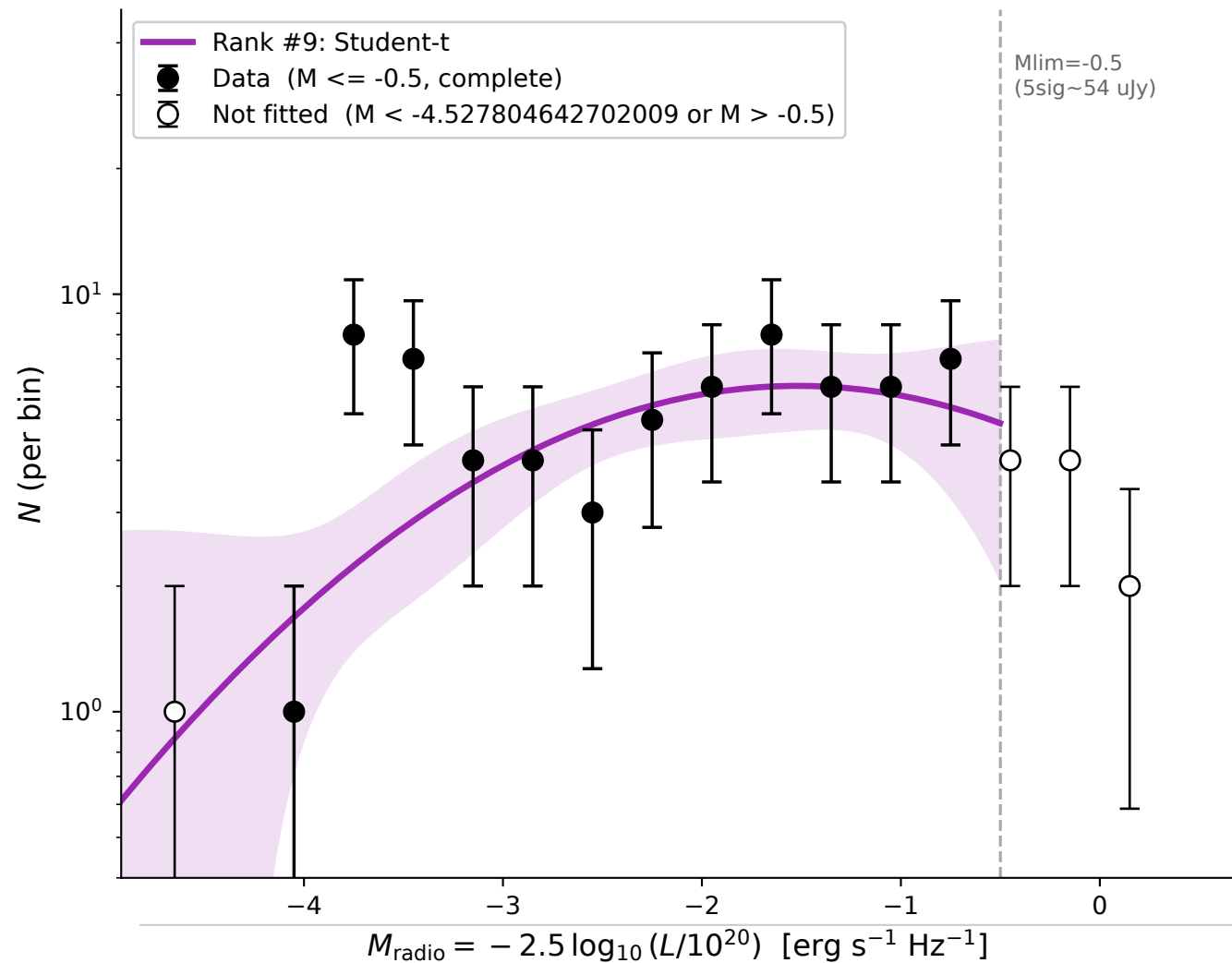
Fitted equation (parameter values substituted):

$N(M) = 19.23 * f_{\text{SN}}(M; \mu=-0.557, \sigma=2.255, \alpha_{\text{sk}}=-30.000)$

Residuals



Rank #9: Student-t



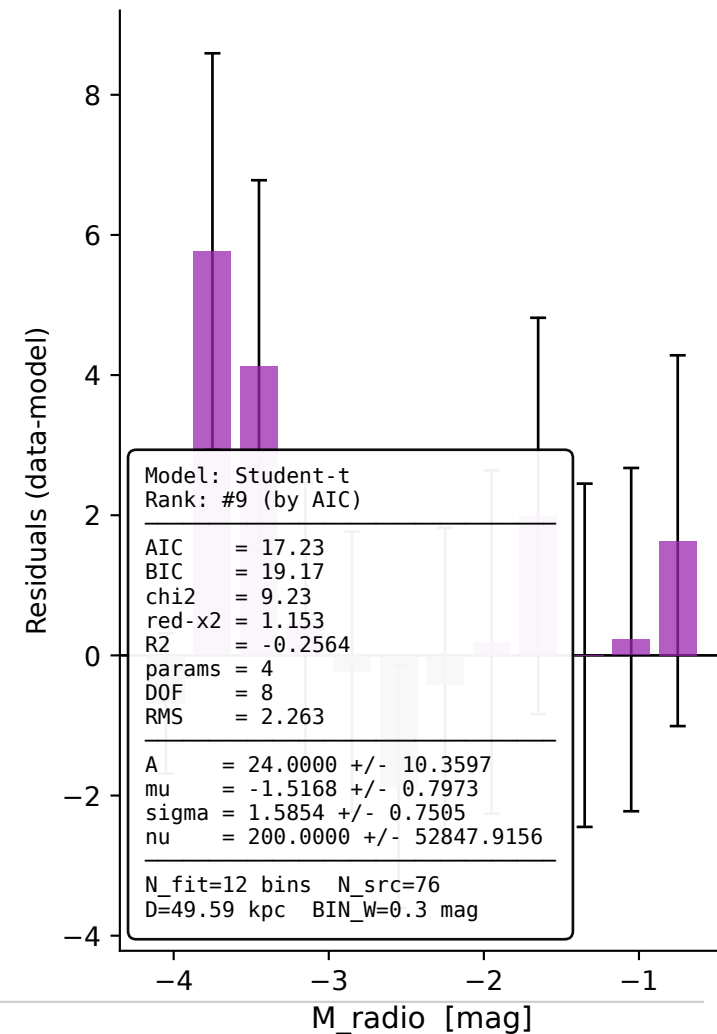
General model equation:

$N(M) = A * t_{\text{nu}}((M - \mu) / \sigma)$
 $t_{\text{nu}}(x)$ proportional to $(1 + x^2/\nu)^{-(\nu+1)/2}$
 A = amplitude, μ = centre, σ = scale, ν = degrees of freedom

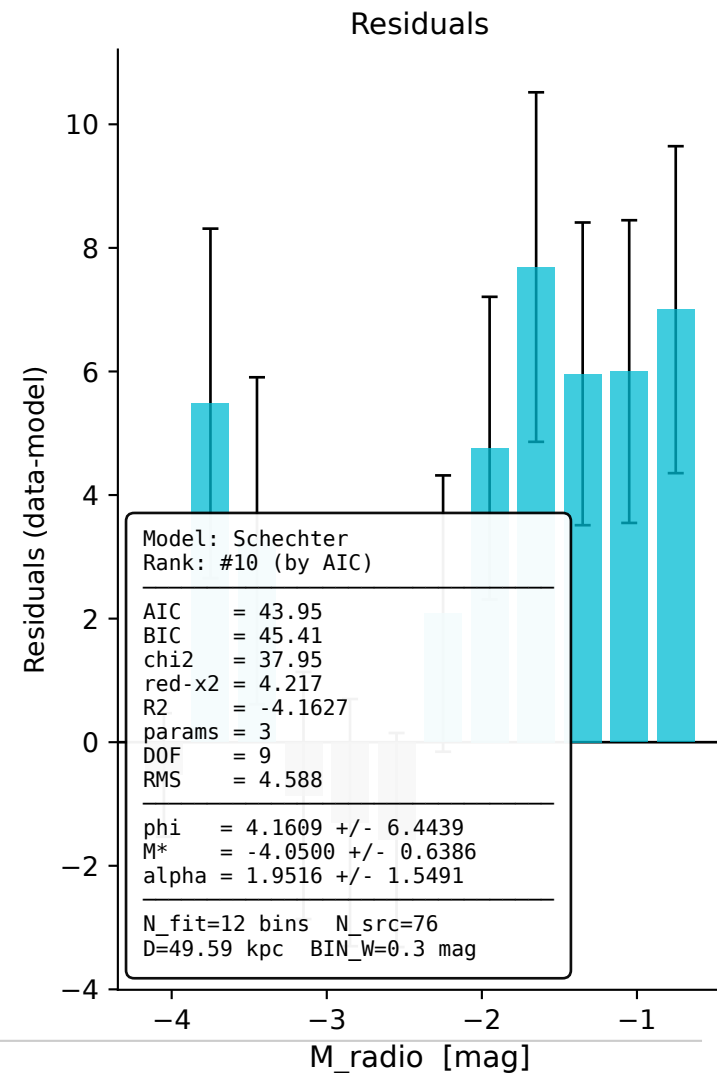
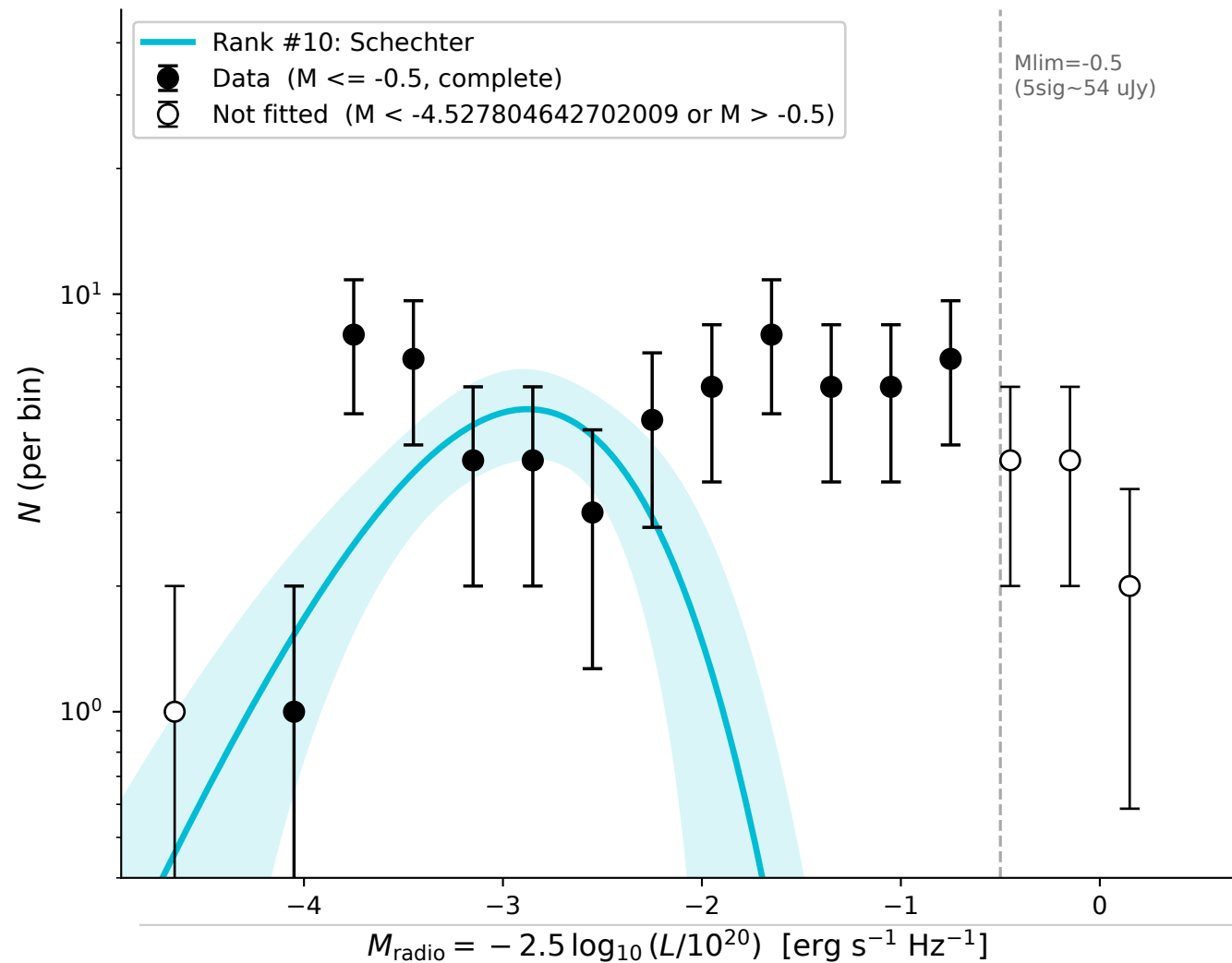
Fitted equation (parameter values substituted):

$N(M) = 24.00 * t_{200.0}((M - (-1.517)) / 1.585)$

Residuals



Rank #10: Schechter



General model equation:

$$N(M) = \phi * 10^{(0.4 * (\alpha + 1) * (M - M^*))} * \exp[-10^{(0.4 * (M - M^*))}]$$

 ϕ = normalisation, M^* = bright cutoff, α = faint-end slope
 Originally for galaxy luminosity functions (Schechter 1976)

Fitted equation (parameter values substituted):

$$N(M) = 4.161 * 10^{(0.4 * (1.952 + 1) * (M - (-4.050)))}$$

$$* \exp[-10^{(0.4 * (M - (-4.050)))}]$$